

Python – Modular Programming & Functions MCQs

Answer Key and Study Guide (First 20 Questions)

1. Which keyword is used to define a function in Python?

- ☐ A) function
- ☐ B) define
- ☐ C) def
- ☐ D) fun

Answer: C) def

2. What is the main purpose of modular programming?

- ☐ A) To make programs longer
- ☐ B) To divide a program into smaller, reusable components
- ☐ C) To avoid using variables
- ☐ D) To eliminate functions

Answer: B) To divide a program into smaller, reusable components

3. Which of the following correctly defines a function?

- ☐ A) function add(a,b):
- ☐ B) def add(a,b):
- ☐ C) define add(a,b):
- ☐ D) func add(a,b):

Answer: B) def add(a,b):

4. What is the output of the following code?

```
def square(x):  
    return x * x  
print(square(5))
```

- A) 10
- B) 20
- C) 25
- D) 125

Answer: C) 25

5. What is the purpose of the return statement in a function?

- ☐ A) To print the function

- o B) To terminate Python
- o C) To send a value back to the caller
- o D) To define a parameter

Answer: C) To send a value back to the caller

6. What is the output?

```
def add(a, b):  
    return a + b  
x = add(10, 20)  
print(x)
```

- A) 10
- B) 20
- C) 30
- D) None

Answer: C) 30

7. Which of the following is a parameter?

```
def multiply(x, y):  
    return x * y
```

- A) multiply
- B) x and y
- C) return
- D) x * y

Answer: B) x and y

8. In the following statement, what are 10 and 20?

```
add(10, 20)
```

- A) Parameters
- B) Arguments
- C) Functions
- D) Variables only

Answer: B) Arguments

9. What happens if a function does not contain a return statement?

- o A) It always returns 0
- o B) It returns None
- o C) It produces a syntax error
- o D) It returns False

Answer: B) It returns None

10. What is the output?

```
def test():  
    x = 10  
    test()  
    print(x)
```

- A) 10
- B) None
- C) NameError
- D) 0

Answer: C) NameError

11. The variable x defined inside a function is generally:

- A) A global variable
- B) A local variable
- C) A class variable
- D) A constant

Answer: B) A local variable

12. Which is an advantage of using functions?

- A) Code reuse
- B) Better organization
- C) Easier testing and debugging
- D) All of the above

Answer: D) All of the above

13. What is the output?

```
def greet(name="Student"):  
    return "Hello " + name  
print(greet())
```

- A) Hello
- B) Hello Student
- C) Student
- D) Error

Answer: B) Hello Student

14. Which function definition uses a default parameter?

- A) `def add(a, b):`
- B) `def add(a, b=10):`
- C) `def add(a=10, b):`
- D) `def add(a, b, c):`

Answer: B) `def add(a, b=10):`

15. What is the output?

```
def calculate(a, b):
    sum_value = a + b
    product = a * b
    return sum_value, product
x, y = calculate(3, 4)
print(x, y)
```

- A) 7 12
- B) 12 7
- C) 7
- D) Error

Answer: A) 7 12

16. Which of the following best represents modular design?

- A) Writing the entire program inside one large function
- B) Dividing a program into functions such as `read_data()`, `process_data()`, and `display_result()`
- C) Avoiding functions
- D) Using only global variables

Answer: B) Dividing a program into functions such as `read_data()`, `process_data()`, and `display_result()`

17. Consider:

```
def preprocess(data):
    data = data / 10
    return data
def calculate(data):
    return data * 5
x = preprocess(20)
y = calculate(x)
print(y)
```

- A) 2
- B) 10
- C) 20
- D) 100

Answer: B) 10

18. Which approach is most appropriate for writing a modular data-processing program?

- A) One large block containing all operations
- B) Separate functions for input, preprocessing, analysis, and output
- C) Only global variables
- D) Repeating the same code multiple times

Answer: B) Separate functions for input, preprocessing, analysis, and output

19. What is the output?

```
def modify(x):  
    x = x + 10  
a = 5  
modify(a)  
print(a)
```

- A) 5
- B) 10
- C) 15
- D) Error

Answer: A) 5

20. Which of the following improves function reusability?

- A) Hard-coding values inside the function
- B) Using appropriate parameters
- C) Using many global variables
- D) Duplicating the function

Answer: B) Using appropriate parameters